

If I Invest *Tomorrow*

A financial decision simulator with an autonomous AI agent

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Indian equities, gold, bonds and cash · data to 1 October 2026 · demo money only

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Agenda

01 PROBLEM

Question, objectives,
what was delivered

02 SYSTEM

Architecture and data

03 METHODS

Features, regimes, ML, CAPM and
risk,
optimisation, Monte Carlo

04 RESULTS

Tomorrow's plan, probabilities,
stress tests, timing, backtest

05 AGENT & APP

Qwen3.5-4B agent, autonomous
trading, web application

06 CLOSE

Verification, limitations,
future work, conclusion

The question we answer

Not “*which stock will rise?*” but:

1

What should I invest in tomorrow?

2

How much should go into each asset, in rupees and shares?

3

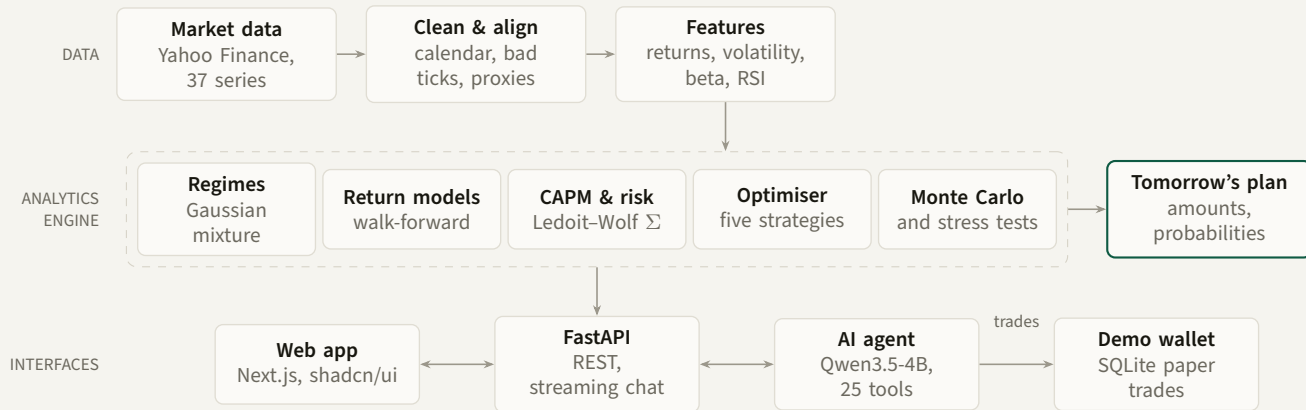
What is the **probability** of reaching my target, and what happens in a crash?

Most prediction tools give a single expected price. Investors need the *range* of outcomes, the risk they take, and a concrete, executable plan that fits their amount, horizon (1/3/5 years), risk tolerance, target return and preferred sectors.

Objectives and what was delivered

OBJECTIVE IN THE PROPOSAL	DELIVERED	STATUS
Capture investor inputs	Amount, horizon, risk, target, sectors — via form <i>or</i> plain-language chat	Done
Analyse market conditions	Returns, volatility, correlation, beta, GMM market regimes, anomaly detection	Done
Financial engineering	CAPM, efficient frontier, Sharpe, VaR, CVaR, max drawdown	Done
AI / data-science estimation	Ridge, Random Forest, XGBoost; walk-forward validated, volatility model beats baselines	Done
Tomorrow's investment plan	Allocation with rupee amounts and whole-share quantities	Done
Monte Carlo + target probability	10,000 regime-switching, fat-tailed paths per strategy	Done
Stress tests	−2/−5/−10% shocks, sector crash, rate and oil shocks, 3 historical replays	Done
Compare five strategies	Max Return, Min Risk, Max Sharpe, Goal-Based, Crash-Resistant	Done
Beyond the proposal	Autonomous AI agent, demo-money trading, web app, Docker, backtest, SIP analysis	Added

Architecture



Docker Compose · the model runs locally in Ollama on an 8 GB GPU

Data: Indian markets, 2014–2026

UNIVERSE	COUNT
NSE stocks across 12 sectors (NIFTY 50 members)	26
Index ETFs: NIFTY 50, Next 50, Bank	3
Gold ETF · G-Sec bond ETF · liquid fund	3
Benchmark: NIFTY 50 index	1
Macro: India VIX, Brent, USD/INR, US 10Y, Bank NIFTY	5

TRADING DAYS

2,950

INVESTABLE ASSETS

32

OF DAILY OHLCV

12 yrs

DATA PROBLEMS FOUND AND FIXED

- **Wrong-scale prints** on 19–20 Dec 2019 in three ETFs (price $\div 10$ to $\div 100$) — caught by a rolling-median filter.
- **Illiquid gilt ETF**: stale and erratic prints; tighter $\pm 4\%$ band around the 11-day median.
- **Liquid fund**: Yahoo's NAV ignores daily distributions; rebuilt from the RBI repo-rate path.
- All series aligned to the NSE trading calendar; small gaps forward-filled.

Feature engineering

PER ASSET, STRICTLY BACKWARD-LOOKING

- Returns over 1, 5, 21, 63, 126, 252 days
- Realised volatility (21, 63 days) and their ratio
- Price vs 20/50-day averages; 50 vs 200-day
- RSI(14), MACD histogram, 252-day drawdown
- Rolling 63-day beta, correlation and skew vs NIFTY

MARKET STATE (SHARED BY ALL ASSETS)

- NIFTY 21/63-day return, volatility, drawdown, gap to 200-day average
- India VIX level and 21-day change
- Brent, USD/INR and US 10-year moves

TARGET

Forward 21-trading-day return of each asset
panel of 32 assets $\times$ 2,950 days

Market regimes

Gaussian mixture on NIFTY return, volatility, drawdown and India VIX



REGIME	SHARE OF DAYS	RETURN P.A.	VOLATILITY	AVG VIX	PERSISTENCE
■ Bull / Calm	56%	+13.0%	10.8%	13.6	96.3%
■ Neutral / Sideways (today)	42%	+6.5%	17.6%	19.0	94.9%
■ Bear / Volatile	2%	+1.1%	54.5%	44.2	98.5%

Forecasting returns, honestly

Expanding-window walk-forward, 2021–2026, purged targets; 43,400 out-of-sample predictions

MODEL	RMSE	DIRECTION	IC
Ridge regression in use	0.0566	51.2%	0.027
Random Forest	0.0572	50.6%	0.001
XGBoost	0.0583	50.4%	−0.002
Historical mean (baseline)	0.0565	47.5%	—

IC = average daily Spearman correlation between forecast and realised return across assets.

No model beats the naive baseline on error. Cross-sectional skill is at most small (IC 0.00–0.03).

So the forecast is used only as a *relative tilt*, weighted by measured skill:

$$\lambda = \text{clip}(5 \cdot \text{IC}, 0, 0.5) = 13.5\%$$

A model with no skill would get zero weight automatically.

Volatility is forecastable; simulated outcomes are checked against reality

NEXT-MONTH VOLATILITY, WALK-FORWARD 2021–26

MODEL	R^2	ERROR
Ridge on range-based features <i>in use</i>	0.58	25%
Same as last quarter	0.50	29%
Same as last month	0.39	31%

40,423 forecasts, settings chosen on 2019–20 only. Trees and ensembles scored lower. Liquid fund excluded.

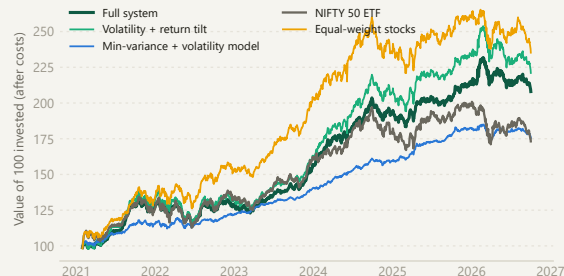
MONTE CARLO CALIBRATION, 1-YEAR HORIZON

- 104 forecasts from quarterly dates 2019–25, information as of each date
- 90% band held the outcome 81% of the time (100% in 2021–23, 44% in 2020)
- The miss is the COVID crash and the rebound; realised returns beat expected by 7% on average
- Reported as measured, not re-tuned to fit

Does the model stack create value? Walk-forward backtest, 2021–26

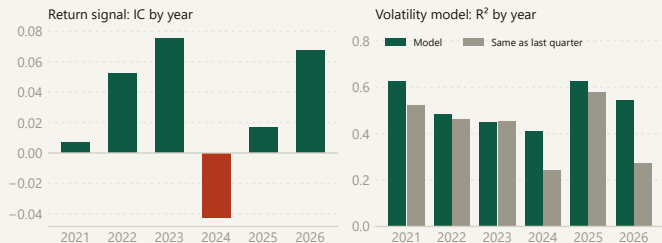
STRATEGY	CAGR	VOL	SHARPE	MAX DD	TURN./YR
Min-variance	11.5%	5.7%	0.96	−6.1%	0.8×
Max-Sharpe, prior	12.8%	11.4%	0.60	−16.1%	3.3×
+ volatility model	14.5%	11.3%	0.75	−15.2%	4.6×
+ 13.5% return tilt	15.3%	12.1%	0.77	−16.5%	4.7×
+ regime overlay	14.1%	10.1%	0.80	−13.6%	5.0×
(full)					
NIFTY 50 ETF	10.4%	12.8%	0.34	−16.1%	0.2×
Equal-weight stocks	16.6%	13.0%	0.82	−15.0%	0.7×

Monthly rebalance, long-only, 0.10% cost per trade; every input uses only data available at each date.



The volatility model and regime overlay add *risk control*; the return tilt adds little. An equal-weight stock basket matches the full system in this one bull-market sample.

Robustness: year by year, calibration, and what regimes and anomalies tell us



Return signal: IC positive in 5 of 6 years, never significant on its own. Volatility model: ahead of the baseline in 5 of 6 years, tied in 2023.

- **Calibration.** Intervals of 50 / 75 / 90 / 95 / 99% held 44 / 69 / 81 / 84 / 93%. Fatter tails or a block bootstrap do not fix it: the miss is the 2020 crash.
- **Regimes.** Calm $\rightarrow$ Neutral in a month 18% of the time. After Neutral, 35% of months saw a $\geq 5\%$ market drawdown against 13% after Calm. They do not improve the volatility forecast (VIX already carries it).
- **Anomalies.** Flagged days are followed by 21–23% volatility vs 13% and a 5% drawdown 60–65% of the time vs 21%, but not by lower returns: a risk warning, not a sell signal.
- **Return signal.** Relative-strength features lift IC to 0.04 (not significant on independent windows): a candidate, not adopted.

Expected returns and the risk model

CAPM

$$\mathbb{E}[R_i] = r_f + \beta_i \text{ERP}, \quad r_f = 6\%$$

ERP blends 5-year NIFTY history with a 5.5% prior, clipped to [3%, 8%] $\Rightarrow$ 3.0%, market return 9.0%.

FINAL EXPECTED RETURN

$$\mu_i = 0.6 \text{CAPM}_i + 0.4 \text{History}_i + \lambda \cdot \text{Tilt}_i^{\text{ML}}$$

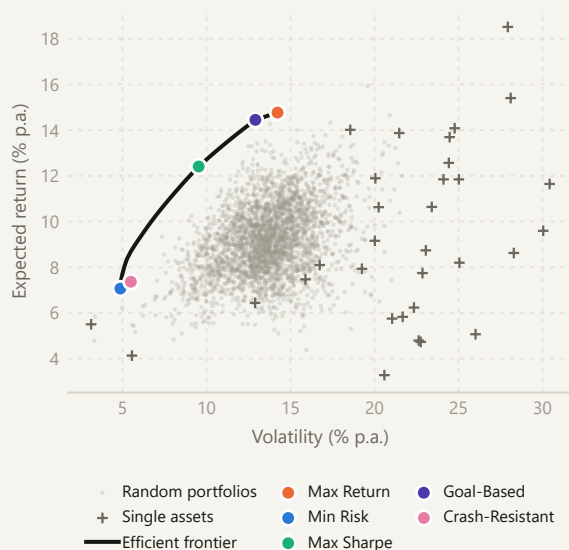
RISK

- Covariance Σ : Ledoit–Wolf shrinkage on 5 years of daily returns
- VaR_{95} and $\text{CVaR}_{95} = \mathbb{E}[L \mid L \geq \text{VaR}]$
- Maximum drawdown, Sharpe $= (\mu - r_f)/\sigma$
- Oil and rate sensitivities for stress tests
- Isolation Forest flags anomalous market days

Five strategies on one efficient frontier

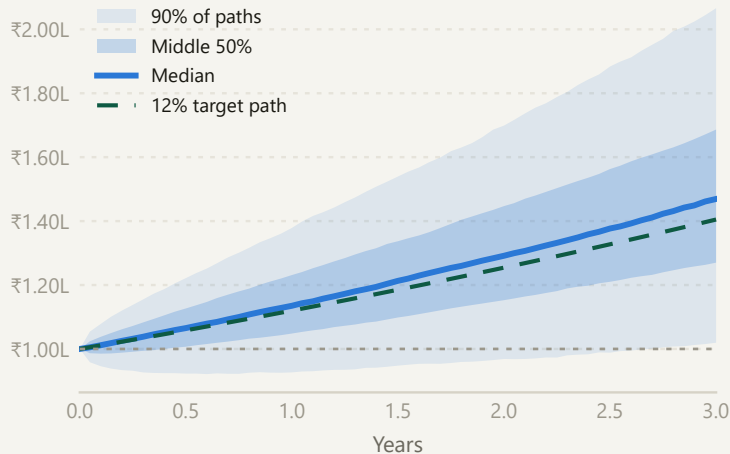
STRATEGY	OPTIMISATION
Max Return	$\max_w w^\top \mu$ (linear program)
Min Risk	$\min_w w^\top \Sigma w$
Max Sharpe	$\max_w (w^\top \mu - r_f) / \sqrt{w^\top \Sigma w}$
Goal-Based	frontier point with the highest <i>simulated</i> P(reach target)
Crash-Resistant	$\min \text{CVaR}_{95}$ over history <i>and</i> stress scenarios (Rockafellar–Uryasev LP)

Long-only, fully invested; per-asset caps by risk profile (single stock 8 / 15 / 25%); preferred sectors restrict the stock universe.



Monte Carlo engine

- **10,000 daily paths** per strategy over the horizon
- **Fat tails:** Student- t innovations, $\nu = 5$
- **Regime switching:** 3-state Markov chain from the GMM transition matrix, started from today's regime
- Drift $\ln(1 + \mu) - \frac{1}{2}\sigma_k^2$ so the mean equals the expected return
- **Day-0 shocks** move the market into the bear regime
- **Outputs:** P(gain), P(target), VaR, CVaR, drawdowns, percentile bands



Recommended plan, ₹1,00,000 over 3 years: median, middle 50% and 90% of paths against the 12% target path.

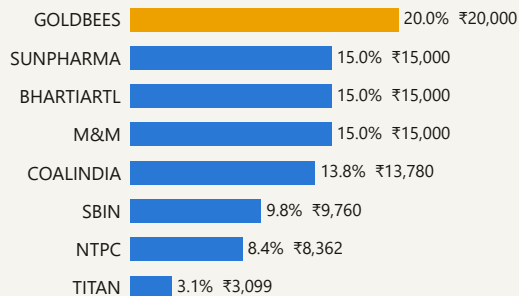
Tomorrow's investment plan

Sample investor: ₹1,00,000 · 3 years · medium risk · 12% target · prices at the close of 1 Oct 2026

Goal-Based

RECOMMENDED

Within medium-risk limits and the best chance (58%) of reaching 12%; ties within 2 pp go to the higher Sharpe.



■ Stocks 76% ■ Gold 20% ■ Cash & residual

EXPECTED RETURN

14.4%

CHANCE OF GAIN

96%

EXPECTED DRAWDOWN

14.5%

VOLATILITY

12.9%

REACH 12% TARGET

58%

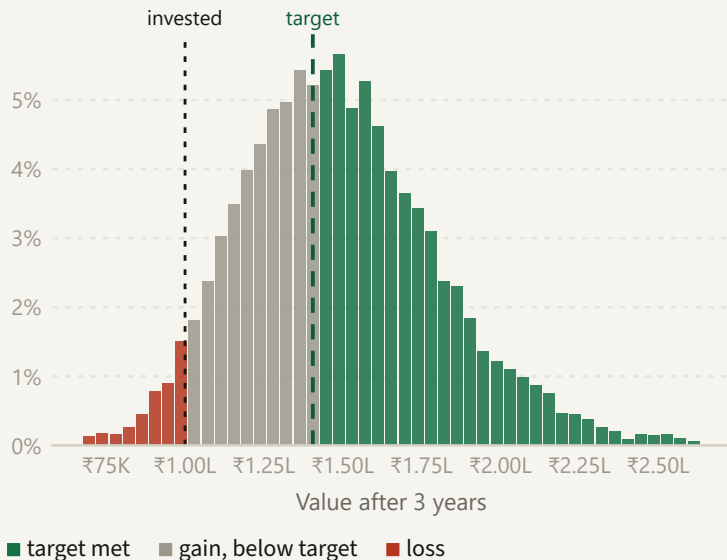
STRESS TESTS PASSED

8 of 9

Sharpe 0.65 · beta 0.74 · 1-year VaR₉₅ 7.3% · CVaR₉₅ 13.9%

Whole shares at the latest close, after 0.1% trading costs.

What could happen: the distribution, not a point



VALUE AFTER 3 YEARS

Worst 5%	₹1,01,941
Median	₹1,46,925
Best 5%	₹2,06,622

PROPOSAL'S ILLUSTRATION VS SYSTEM OUTPUT

	PROPOSAL	SYSTEM
Expected return	~12.4%	14.4%
Volatility	~14.2%	12.9%
P(positive)	~73%	96%
P(target)	~68%	58%

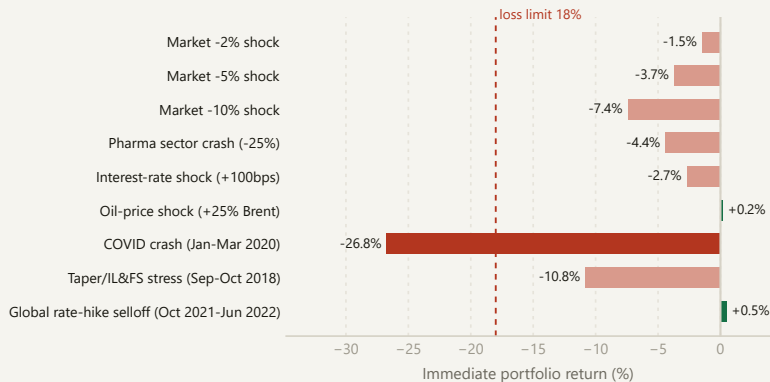
Five strategies compared

Same investor, same 10,000 simulated futures for every strategy

STRATEGY	RETURN	VOLATILITY	SHARPE	CHANCE OF GAIN	REACH TARGET	MEDIAN VALUE
■ Max Return	14.8%	14.2%	0.62	94%	58%	₹1,47,461
■ Min Risk	7.1%	4.9%	0.22	99%	4%	₹1,22,476
■ Max Sharpe	12.4%	9.5%	0.67	98%	50%	₹1,40,605
■ Goal-Based recommended	14.4%	12.9%	0.65	96%	58%	₹1,46,925
■ Crash-Resistant	7.4%	5.5%	0.25	99%	7%	₹1,23,409

Decision rule. Keep strategies inside the investor's limits on volatility, worst stress loss and one-year VaR; pick the highest probability of reaching the target; within 2 percentage points prefer the higher Sharpe ratio. Defensive strategies are almost certain to gain but rarely reach a 12% goal.

Stress tests



MONTE CARLO AFTER A DAY-0 SHOCK

SCENARIO	MEDIAN	P(TGT)
No shock	₹1,47,278	59%
Market -2%	₹1,45,088	56%
Market -5%	₹1,37,454	47%
Market -10%	₹1,32,146	41%
Pharma crash	₹1,36,467	46%
Rate +100 bp	₹1,38,899	48%
Oil +25%	₹1,47,545	60%

Only a COVID-scale crash (**-26.8%**) breaches the 18% medium-risk loss limit.

Invest now, wait, or SIP?

All options run on the same simulated markets; uninvested money earns the risk-free rate

OPTION	MEDIAN AFTER 3 YR	BEATS LUMP SUM
Invest now (lump sum)	₹1,46,205	—
Wait 1 month	₹1,45,444	44%
Wait 3 months	₹1,43,725	38%
Wait 6 months	₹1,41,197	34%
SIP over 6 months	₹1,44,301	37%
SIP over 12 months	₹1,42,120	32%

With a positive expected return, **time in the market beats timing it**: every delay lowers the median outcome.

A SIP is still useful when the investor values a narrower range of bad outcomes over the higher median — it wins in about a third of futures.

RESULTS

Walk-forward backtest

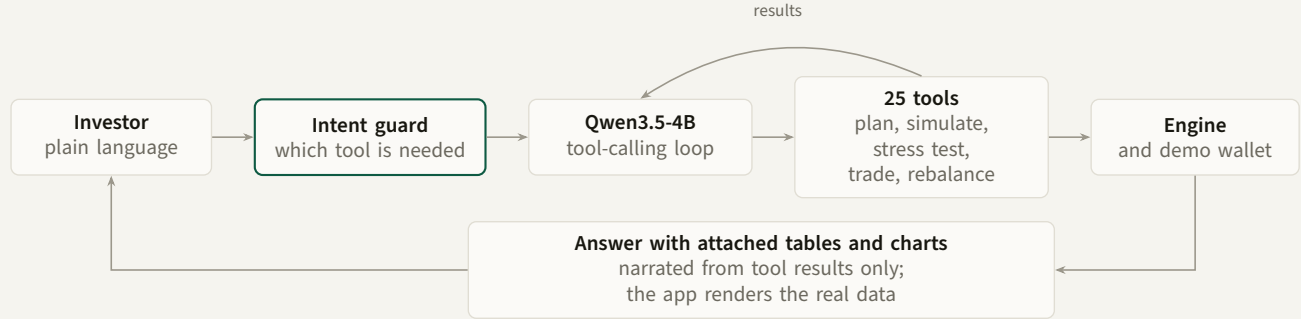
Weights chosen only from data before 3 Oct 2023, rebalanced quarterly, held to 1 Oct 2026; no ML (no look-ahead)



STRATEGY	TOTAL RETURN	CAGR	VOLATILITY	SHARPE	MAX DRAWDOWN
■ Goal-Based	+55.1%	16.1%	8.9%	1.14	−8.1%
■ Max Sharpe	+58.8%	17.1%	10.3%	1.07	−9.3%
■ Min Risk / ■ Crash-Resistant	+39.1% / +38.1%	11.9 / 11.6%	5.9 / 6.0%	1.00 / 0.95	−6.8 / −6.3%
NIFTY 50 benchmark	+14.8%	4.8%	13.3%	−0.09	−15.8%

The agent: Qwen3.5-4B with tools

Runs locally through Ollama; the language model chooses tools and explains, the engine produces every number



GROUNDING

Early tests: the 4B model invented an allocation. Now, if a request needs a tool and the model skips it, the agent runs it and the model only narrates.

GUARDS IN CODE

A strategy sticks only if the user named it; trades need an instruction, not a question; resets need an explicit “reset”.

MEASURED

23 / 23 scripted conversation cases pass, covering analysis, trading, stop-loss handling and ask-first confirmation.

Autonomous trading with demo money

TWO MODES

Auto-trade — “invest it”, “sell all my TCS”, “rebalance” execute immediately. *Questions never trade*: “should I sell TCS?” gets analysis.

Ask first — every order is staged; it executes only after the investor’s own “confirm”.

DEMO WALLET

₹10,00,000 of virtual cash in SQLite; fills at the latest close with 0.05% brokerage + 0.05% slippage; atomic multi-order execution.

HEALTH CHECK & AUTO-MANAGE

Stop-loss	position down 15% from cost $\Rightarrow$ sell
Drift	holding above its cap by 3 pp $\Rightarrow$ trim
Risk limit	volatility $> 1.1 \times$ limit $\Rightarrow$ de-risk
Bear regime	equity $> 60\% \Rightarrow$ move to Crash-Resistant
Idle cash	$> 25\%$ of wallet $\Rightarrow$ deploy

Autopilot runs this review when switched on and after every market-data refresh.

Web application: the planner

Next.js 16, TypeScript, Tailwind CSS, shadcn/ui · screenshot of the running app (data refreshed 2 Oct 2026)

If I Invest Tomorrow

OverviewAgentPlannerMarketsPortfolio

Auto-trade ₹11,99,615
▼ -0.03%

NIFTY 50 22,620.45 ▼ 0.42% BANK NIFTY 54,633 INDIA VIX 13.49 USD/INR 96.05 BRENT \$103.53 REGIME Neutral / SidewaysClose of 30 Sept 2026 · qwen3.5:4b

PLANNER

Design tomorrow's investment

Five optimised strategies for your profile, each simulated ten thousand times and stress-tested against shocks and past crashes.

AMOUNT

₹ 1,00,000

50K 1L 5L 10L

HORIZON

1 year 3 years 5 years

RISK TOLERANCE

Low Medium High

TARGET RETURN 12.0% a year

Max ReturnMin RiskMax SharpeGoal-Based RECCrash-Resistant

PlanStrategiesMonte CarloStress testsNow, wait or SIPBacktest

TOMORROW'S PLAN · ₹1,00,000 · 3 YR · MEDIUM RISK

Goal-Based RECOMMENDED

Invest this plan

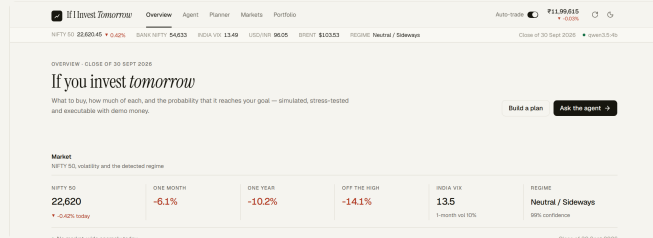
Frontier portfolio with the highest simulated probability of hitting your target return.

EXPECTED RETURN	VOLATILITY	CHANCE OF GAIN	12% TARGET	DRAWDOWN	STRESS TESTS
14.1%	12.2%	96%	56%	13.6%	8 of 9
per year	per year	over 3 yr	10,000 simulations	1 in 20: 26%	Sharpe 0.67 · β 0.69

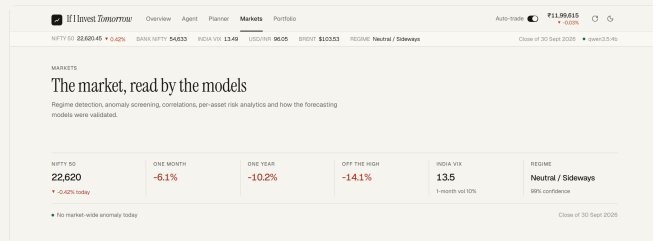
Profile on the left; the five strategies; the recommended plan with its key figures, risk-limit check, allocation and composition. Tabs open the Monte Carlo, stress tests, timing and backtest views.

Web application: overview, agent, markets, portfolio

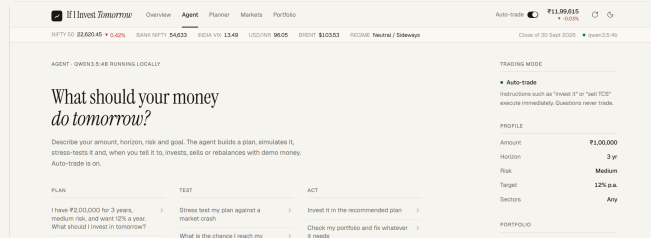
OVERVIEW



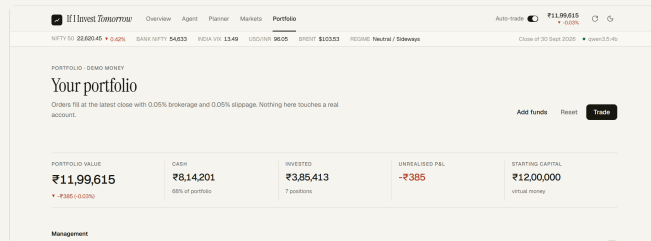
MARKETS



AGENT



PORTFOLIO



Engineering and verification

STACK

Python 3.13 · pandas, scikit-learn,
XGBoost, PyTorch, SciPy

FastAPI with streaming chat

Ollama + Qwen3.5-4B (RTX 4070, 8 GB)

Next.js 16, TypeScript, Tailwind,
shadcn/ui, Recharts

Docker Compose (API + web)

UNIT TESTS PASSING

55

API ENDPOINT CHECKS

27

AGENT CONVERSATION CASES

23 / 23

WHAT THE TESTS COVER

CAPM, VaR/CVaR and drawdown maths;
optimiser constraints for every risk level;
Monte Carlo mean and shock behaviour;
wallet accounting; trade intent detection;
confirmation and stop-loss guards.

Code: github.com/Tendool/If-I-invest-tomorrow

Limitations

- **Weak predictability:** monthly returns are close to unforecastable; the ML adds only a small tilt.
- **Assumptions:** risk-free rate 6%, ERP prior, sector rate sensitivities and bond duration are set by hand.
- **One backtest window:** NIFTY was weak over 2023–26 in this data, which flatters diversified portfolios.
- **Data:** Yahoo Finance prices; the liquid fund is a reconstructed proxy; no taxes (STT, capital gains).
- **Small language model:** occasional slips in wording; the tables under each answer are authoritative.
- **Simulation only:** demo money, not investment advice.

Future work

SIGNALS

News sentiment with FinBERT as a model feature; macro factors from RBI data; Black-Litterman views instead of a fixed blend.

REALISM

Indian tax and transaction costs; mutual funds and SIP products; multi-period rebalancing; intraday data.

AGENT

Benchmark 7-9B local models on the same 23 cases; scheduled autopilot; explanation of each trade in the history.

Conclusion

From “*what might happen*” to “*what to do tomorrow*”.

INTEGRATED PIPELINE

1

Data, ML, CAPM, optimisation, Monte Carlo and stress testing in one system.

STRATEGIES, ONE RULE

5

Compared on the same futures and chosen transparently against the investor's limits.

AGENT THAT ACTS

1

Plans, simulates, invests, sells and rebalances — grounded in the engine, guarded in code.

The honest finding matters as much as the plan: forecasting single-month returns barely beats a naive baseline, so the value of the system lies in measuring risk, comparing strategies and showing the full range of outcomes.

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Thank *you*

Questions and live demo

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github.com/Tendool/If-I-invest-tomorrow